workable-items — Operator Guide

Revision: 1 **Last modified:** 2026-05-28T17:00:00Z **Authority:** vasic-digital tmux project **Maintainer:** milosvasic **Scope:** Operator-facing guide for the v1.0.15 project-local `workable-items` Go binary — the §11.4.93 SQLite Single-Source-of-Truth for every workable item tracked in `Issues.md` / `Fixed.md`.

---

1. What this is

`cmd/workable-items/` is a small Go binary that owns the canonical form of every workable item in the tmux project. The on-disk truth is `docs/workable_items.db` — a SQLite file **tracked in git per §11.4.95**. Markdown (`Issues.md` / `Fixed.md`) is regenerated FROM the DB; the DB is regenerated FROM the Markdown. Round-trip drift is detected by `workable-items diff`.

Two design constraints govern the layout:

- **§11.4.93 SQLite-SSoT mandate.** Workable items must live in a structured, queryable store — not free-form Markdown — so that validation (§11.4.33 closure vocabulary, §11.4.91 description clarity, §11.4.5 / §11.4.69 evidence-required-on-close) can be enforced mechanically.
- **§11.4.95 DB tracked in git mandate.** `docs/workable_items.db` is checked into git on every commit. Operators inspecting the tracker get a deterministic snapshot per SHA; there is no "DB on the server, Markdown in git" split-brain.

The constitution submodule (`constitution/scripts/workable-items/`) ships a Phase-2 scaffold (stubs + canonical `schema.sql`). The Phase 3+ implementation lives in THIS repo at `cmd/workable-items/` per the `feedback_no_modify_constitution` memory rule. Upstream PR to HelixDevelopment/HelixConstitution is tracked as a follow-up.

---

2. Quick start

```
# Build (no CGO, no system SQLite required)
go build ./cmd/workable-items
# → produces ./workable-items in CWD

# Show every item in the DB
./cmd/workable-items/workable-items report

# Validate the DB against §11.4 mandates
./cmd/workable-items/workable-items validate
```

```
# Detect Markdown ↔ DB drift (exits 1 on drift)
./cmd/workable-items/workable-items diff

# Sync DB → regenerated Markdown (writes under docs/workable-
        items/regen/)
./cmd/workable-items/workable-items sync db-to-md

# Sync Markdown → DB (re-parse Issues.md + Fixed.md, allocate ATM-
        NNNs)
./cmd/workable-items/workable-items sync md-to-db

# Add a new item
./cmd/workable-items/workable-items add \
  --type Bug --severity HIGH \
  --title "Wheel scroll broken on Termux" \
  --description "Pinch-to-zoom intercepts touch before tmux sees
        wheel events; need explicit binding for Termux's gesture
        handler" \
  --category A

# Close an item with evidence (§11.4.5 + §11.4.69)
./cmd/workable-items/workable-items close ATM-042 \
  --status fixed \
  --evidence qa-results/runs/2026-05-28T17-00-00Z/test_44.log \
  --by AI --on 2026-05-28
```

Run `workable-items --help` for the full subcommand reference.

---

3. Where things live

Path	Role
`cmd/workable-items/`	Go source (11 files: model / parser / db / sync_md_to_db / sync_db_to_md / diff / validate / add / close / main + tests)
`cmd/workable-items/schema.sql`	Embedded schema — drift-checked verbatim copy of `constitution/scripts/workable-items/schema.sql@6828ff2`
`cmd/workable-items/testdata/`	Golden corpus (`golden_issues.md` / `golden_fixed.md` + exports) for round-trip byte-identical tests
`docs/workable_items.db`	The canonical DB — **TRACKED in git per §11.4.95**
`docs/workable-items/Status.md`	Project-local Status document (§11.4.45)
`docs/workable-items/Status_Summary.md`	Two-audience companion (§11.4.56)
`docs/workable-items/regen/`	Output of `sync db-to-md` (gitignored — re-generate on demand)

---

4. Subcommands

Subcommand	Purpose
`sync md-to-db`	Parse `Issues.md` + `Fixed.md`, upsert rows in the DB, allocate ATM-NNN IDs for new items. Idempotent: a second invocation with no Markdown changes allocates 0 new IDs.
`sync db-to-md`	Render the DB back to Markdown under `--out-dir`. Byte-identical to source for the golden testdata corpus; lossy for live free-form bodies (see §6).
`diff`	Open a temp DB, sync `Issues.md` + `Fixed.md` into it, compare against the live DB. Exits 1 on drift unless `--allow-drift`.
`validate`	Walk the DB and report violations: §11.4.33 type-aware closure vocabulary, §11.4.91 description-clarity floor (≥40 chars or ≥6 words), §11.4.5 + §11.4.69 evidence-required for closed items. Exits 1 if any findings.
`add`	Append a new item. ATM-NNN allocated monotonically. Required: `--type`, `--severity`, `--title`, `--description`.
`close`	Mark an item closed. Required: `--status fixed|implemented|completed|obsolete`, `--evidence PATH`. Per §11.4.90, `obsolete` additionally requires `--superseding-item` and `--triple-check-evidence`.
`report`	List items, optionally filtered by `--type`, `--status`, or `--obsolete-audit`.

5. Why a Go binary (not a shell script)

- **No CGO.** Uses `modernc.org/sqlite` — a pure-Go port of SQLite 3.53.1 (v1.51.0 as of 2026-05-28). Cross-compiles on Mistborn (Darwin arm64) and nezha (Linux x86_64) with no build-host SQLite dependency.
- **Atomic writes via WAL.** `OpenDB` enables WAL mode; `Close` runs `PRAGMA wal_checkpoint(TRUNCATE)` so the on-disk file is always self-contained. Compatible with `git add` (no -wal / -shm files left behind for git to choke on).
- **Embedded schema.** `cmd/workable-items/schema.sql` is `//go:embed`-ed into the binary. The DB applies the embedded schema on `OpenDB` if not already applied. A drift-check header on line 1 records the upstream constitution SHA the schema was copied from.
- **`go test ./cmd/workable-items/... -count=3`** runs deterministically per §11.4.50 — 10 tests × 3 iterations = 30 PASS.

6. Honest gaps (§11.4.6 — no guessing)

The following are KNOWN gaps, tracked transparently. None of them is a §11.4 PASS-bluff: each is a documented topology / scope limitation with a follow-up plan.

1. **Live-corpus round-trip is NOT byte-identical.** `Issues.md` / `Fixed.md` carry free-form bodies (forensic anchors, multi-paragraph "Source-side fix" + "Captured evidence" sections, blockquotes, code fences) that cannot be losslessly reconstructed from the structured `items` schema's flat `description` field. The §11.4.93 phase-6 migration plan addresses this; until then, round-trip byte-identical equivalence holds for the **golden testdata corpus only**, and `sync db-to-md` against live data produces simplified bodies under `docs/workable-items/regen/` that are intentionally not used to overwrite the canonical `Issues.md` / `Fixed.md`.

2. **45 legacy items default to `Type=Task`.** The tmux corpus predates §11.4.16 — no `**Type:**` lines exist in the original Issues / Fixed bodies. `workable-items validate` therefore reports §11.4.33 violations for every closed item that used the `RESOLVED` heading hint (mapped to `Fixed (→ Fixed.md)`, the Bug closure word) while the row's `type` is `Task` (which would expect `Completed (→ Fixed.md)`). This is the **expected initial state** of the post-population DB, not a bug in the binary. PWU-Q5 (legacy-items Type=Bug cleanup) is the closure path.

3. **Phase 3+ implementation is project-local.** Per the `feedback_no_modify_constitution` memory rule (no project-local pushes into the constitution submodule), this implementation lives at `cmd/workable-items/`. Upstream PR to HelixDevelopment/HelixConstitution is operator-blocked pending explicit authorisation. The constitution's Phase-2 scaffold and the project's Phase-3+ implementation share the schema verbatim (drift-checked on `OpenDB`).

4. **`commit_all.sh` + `sync_issues_docs.sh` integration is owned by PWU-Q4** (parallel cycle, 2026-05-28). Until that PWU lands, operators must invoke `workable-items sync md-to-db` manually after editing `Issues.md` / `Fixed.md` and re-commit the updated `docs/workable_items.db`.

---

7. Anti-bluff verification

Layer	Mechanism	Where
1 — pre-build gate	`verify.sh` greps for `cmd/workable-items/main.go` + `docs/workable_items.db` present	`scripts/verify.sh`
2 — runtime	`go test ./cmd/workable-items/... -count=3` (30 PASS / 0 FAIL)	`cmd/workable-items/*_test.go`
		`scripts/challenges/tmux.yaml`

Layer	Mechanism	Where
3 — Challenge	`CME-WORKABLE-ITEMS-001` (planned, PWU-Q6)	
4 — paired mutation	M-WORKABLE-ITEMS (planned, PWU-Q6) — strips the §11.4.91 clarity floor and asserts `validate` no longer FAILs on a 1-word description	`scripts/tests/meta_test_false_positive_proof.sh`

---

8. Cross-references

- `Constitution.md` §11.4.93 (SQLite-SSoT), §11.4.95 (DB TRACKED in git), §11.4.33 (type-aware closure), §11.4.91 (description clarity), §11.4.50 (deterministic `-count=3`), §11.4.69 (sink-side evidence applied to `--evidence`)
- `docs/workable-items/Status.md` — live status per §11.4.45
- `docs/workable-items/Status_Summary.md` — two-audience companion per §11.4.56
- `CHANGELOG.md` v1.0.15 — A39 release notes
- Constitution scaffold (Phase 2): `constitution/scripts/workable-items/`

Sources verified 2026-05-28

- **Constitution §11.4.93 / §11.4.95 / §11.4.50 / §11.4.33 / §11.4.69** — `constitution/Constitution.md` @ submodule pointer `6828ff2` (pulled 2026-05-28 per §11.4.37 fetch-before-edit).
- **`modernc.org/sqlite`** — https://pkg.go.dev/modernc.org/sqlite (verified 2026-05-28). Latest published version v1.51.0 (2026-05-28), ships SQLite 3.53.1. Confirmed: pure-Go (CGo-free) driver, import path `modernc.org/sqlite`, driver name `sqlite` for `sql.Open`, WAL + `PRAGMA wal_checkpoint` supported via `_pragma` query param or via direct `PRAGMA` statements.
- **Constitution Phase-2 scaffold** — `constitution/scripts/workable-items/schema.sql` @ `6828ff2`; verbatim drift-checked copy at `cmd/workable-items/schema.sql` header line 1.
- **§11.4.99 sources-verification mandate** — new clause landed in constitution `9e3bcc5` (2026-05-28), short-form mirror in this project's `Constitution.md` / `CLAUDE.md` / `AGENTS.md` / `QWEN.md` (PWU-Q5 owns the propagation row).